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Supplemental Appendix for:
Towards a Treatment Sequencing Strategy: A Systematic Review of Treatment Regimens in Advanced Gastric Cancer/Gastroesophageal Junction Adenocarcinoma
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SUPPLEMENTAL METHODS

Inclusion and Exclusion Criteria

To be included in the analysis, a first-line (1L), second-line (2L), or third-line or beyond (3L+) trial must have included adults 18 years or older with advanced unresectable or metastatic gastric or gastroesophageal junction (GEJ) adenocarcinoma. Prior to receiving 1L therapy, patients could not have received systemic therapy except for neoadjuvant or adjuvant therapy completed one or more years prior to the development of metastasis. Included 2L therapy studies had to treat patients who received a 1L therapy with a platinum- or fluoropyrimidine-containing regimen. Excluded studies were those where patients received only maintenance therapy. Given the finding of Wagner et al (2010) of a statistically significant and conclusive survival benefit for combination chemotherapy versus best supportive care (BSC) in the 1L setting, active therapy versus BSC or placebo only was not included in the current search in the 1L setting. For 1L therapy studies only, at least one arm had to contain a fluoropyrimidine (e.g., fluorouracil [5-FU], capecitabine, S-1, or floxuridine) plus a platinum agent. For 2L therapy studies only, an active therapy could be compared with either BSC or placebo as the control arm. Studies must have reported one or more of the following efficacy endpoints: progression-free survival (PFS), overall survival (OS), objective response rate (ORR), or adverse events (AEs). For inclusion, aside from systematic reviews and meta-analyses, study designs had to be phase 2 or 3 randomized controlled trials (RCTs) (including open-label studies) where the sample size was ≥200 in the 1L or ≥40 in the 2L or 3L or beyond therapy studies. If a study included a mixed population of some patients with gastric or GEJ adenocarcinoma and some patients with a different cancer type, the number of included patients with gastric or GEJ adenocarcinoma had to meet the above sample size criteria. Lastly, systematic reviews and meta-analyses from 2015 to 2016 were used only to identify primary studies that may have been missed in the electronic database searches.

Study Selection Process

There were two phases of screening. Titles and abstracts were screened by two independent researchers (Level 1) to determine eligibility according to the above inclusion/exclusion criteria. Full-text versions of studies selected in Level 1 were screened (Level 2). Disagreements arising at either screening level were resolved with a third researcher to reach consensus.

Data Extraction

Data were extracted for each clinical trial from full-text articles, and grouped by line of therapy. Data items included citation/year of publication, trial identification and acronym, population demographics and disease characteristics, treatments and dosing/schedules, trial design, study sites/locations, participant eligibility criteria, method of randomization and blinding, efficacy endpoints (PFS, OS, ORR, in addition to any other reported endpoints), AEs (rates, grades, discontinuations due to AEs or death), and duration of follow-up.

Risk of Bias

The quality of each eligible study was evaluated using the bias assessment tool detailed in Table S4**.** Overall, there was one 1L study with low risk of bias [1], while four and 17 1L RCTs were considered to have moderate to high risk of bias, respectively. Bias was unclear based on the assessment in four 1L studies. Of the 2L studies included, four were deemed to have moderate risk and 20 were considered high risk. Bias risk was unclear in eight 2L studies. Of the 3L+ studies, one had low risk, two were considered to have moderate risk and one was deemed to have high risk. Bias was unclear in two 3L+ studies. Overall, the risk for bias assessment indicated the majority of included 1L and post-1L studies had moderate risk of bias (Table S4).

Description of Included Studies

Study Characteristics

An overview of studies for each line of treatment is detailed in Figure 2**.** Of the 27 1L RCTs identified, one trial was a three-arm study [2], and the remaining 26 were two-arm studies. Twenty-three studies were phase 3, one was phase 2 [2], and one was a pilot study [3]. Four studies [4–7] evaluated patients whose tumors had HER2 overexpression and/or amplification. Two studies [8, 9] included patients with mesenchymal-epithelial transition-positive (MET-positive) tumors. Twelve studies were multiregional; 10 studies were conducted in Asia only, and two were conducted in Europe only. The sample sizes ranged from 202 [10] to 1,053 patients [11], with 13 studies including over 500 patients. Twenty-nine of the 32 unique 2L RCTs were two-arm studies, and three were three-arm studies [12–14]. Fifteen were phase 2 studies, 15 were phase 3 studies, and two were phase 2/3 studies. Nine studies were multiregional; 20 studies were conducted in Asia only, and three were conducted in Europe only. The 2L studies ranged in size from 40 [15] to 741 [14] patients.

Of the six unique RCTs assessing 3L+ treatments, four were phase 3 studies, one was a phase 2 study, and one was a phase 1/2 study. Four were two-arm studies [16–19] and two were three-arm studies [20, 21]. Three studies were multinational, and three studies were conducted in Asia only [17, 19, 21]. The sample sizes of included 3L+ studies ranged from 141 [20] to 507 [18].

Interventions

Figure 3 provides a summary of the treatment interventions for each line of therapy. Ten of the 1L studies assessed the efficacy of a targeted therapy versus chemotherapy or placebo; one study assessed two different doses of the same targeted therapy [6] (**Figure 3a**). Targeted therapies included bevacizumab, cetuximab, lapatinib, onartuzumab, panitumumab, rilotumumab, trastuzumab, pertuzumab, and ramucirumab. All 1L studies included combination chemotherapy in both arms: agents used included capecitabine, cisplatin, docetaxel, epirubicin, oxaliplatin, paclitaxel, S-1, tegafur, 5-fluorouracil (5-FU), and leucovorin (folinic acid). Eleven studies compared the efficacy of chemotherapy doublets [1, 3, 11,

22–29]. Nine studies assessed the efficacy of adding a targeted therapy or simvastatin to first-line chemotherapy doublets [4–6, 9, 30–34]. Three studies compared chemotherapy doublets versus triplets [35–37]; one study compared doublet chemotherapy with two triplet chemotherapy regimens [2]; two studies assessed the efficacy of adding targeted therapy (rilotumumab or panitumumab) to triplet chemotherapy [8, 38]; one study assessed the efficacy of adding a second targeted therapy (pertuzumab) to a targeted therapy plus chemotherapy regimen [7].

Of the 2L studies, 13 assessed the efficacy of a biologic therapy versus chemotherapy or placebo; biologic molecularly targeted or immuno-oncology therapies assessed included nimotuzumab, lapatinib, sunitinib, olaparib, ramucirumab, ipilimumab, pembrolizumab, trastuzumab emtansine, everolimus, and regorafenib (**Figure 4b**). Twelve of the included RCTs were single-agent studies, with a single agent in all treatment arms [12, 14, 17, 39–47]. Twenty-three studies included single chemotherapy agents in one treatment arm. Chemotherapy agents used in the identified 2L studies included docetaxel, irinotecan, paclitaxel, cisplatin, and oxaliplatin. Six studies compared the efficacy of single-agent chemotherapy [12, 14, 40, 41, 46, 48], one study compared single-agent chemotherapy with doublet chemotherapy [49], and one study compared doublet chemotherapy plus targeted therapy with doublet chemotherapy plus placebo [50]. Three studies compared single-single agent chemotherapy with BSC [15, 51, 52] and one study compared dose-escalated paclitaxel with standard dose paclitaxel [47]. Five studies compared doublet chemotherapy with single-agent chemotherapy [53–57]; and one study compared single-agent chemotherapy with single-agent chemotherapy plus valproic acid [58]. Two studies compared biologic therapy with placebo [39, 59]; two studies compared targeted therapy with a taxane [43, 44]; one study compared immunotherapy with BSC [42]; and two studies compared targeted therapy plus BSC with placebo plus BSC [60, 61]. Five studies assessed the efficacy of adding a targeted therapy to single-agent chemotherapy [62–66]. One study compared a tyrosine kinase inhibitor (AZD4547) with paclitaxel [45], and one study assessed re-

challenge of S-1 or cisplatin in combination with docetaxel following progression on cisplatin plus S-1 or cisplatin plus capecitabine [13].

Six unique 3L+ RCTs were identified (**Figure 3c**). JAVELIN Gastric 300 study compared avelumab plus BSC with chemotherapy agents: irinotecan or paclitaxel plus BSC [16], TAGS study compared trifluridine/tipiracil plus BSC with placebo plus BSC [18], CheckMate-032 study compared nivolumab monotherapy and different doses of nivolumab plus ipilimumab [20], ATTRACTION-2 study compared nivolumab with placebo [17], Li et al, 2013 compared two doses of apatinib with placebo, [21] and Li et al, 2016 compared single-dose apatinib with placebo [19].

Patient Characteristics

Patient demographics and disease characteristics are summarized in **Figure 2 and Table S1.**

Median ages reported in the treatment arms of the 1L studies were between 53 and 73 years with the proportion of males ranging from 49% to 83%. Of the 12 studies reporting peritoneal or liver metastases at baseline, across the treatment groups in the included studies, rates of peritoneal metastases ranged from 2.5% to 44% and rates of liver metastases ranged from 22.5% to 52%.

The median ages reported in the treatment arms of the included 2L studies were between 52 and 68 years, and the proportion of males ranged from 47% to 89%. Twenty-two studies reported the incidence of peritoneal or liver metastases at baseline; across treatment groups in the included studies, rates of peritoneal metastases ranged between 19.5% and 56%, and rates of liver metastases ranged between 18% and 61%.

In the treatment arms of the included 3L+ studies, the medians reported for age were between 53 and 66 years with the proportion of males ranging from 48.3% to 87.0%. Three studies reported the incidence of peritoneal or liver metastases at baseline; across treatment groups in the included studies, rates of

peritoneal metastases ranged between 1.5% and 31%, and rates of liver metastases ranged between 17% and 55%.

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